

Gaussian Differential Privacy

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DPRG PRESENTATION

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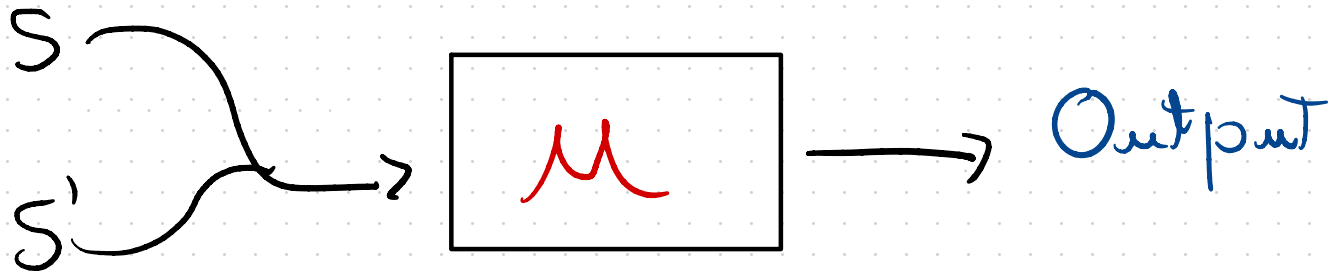
Classical (ϵ, δ) -DP definition:

$$\underline{IP(M(S) \in E) \leq e^\epsilon \cdot IP(M(S') \in E) + \delta}$$

- S and S' are neighboring data bases
- Randomness comes from M
 - ↳ Worst-case over S, S' and E .

How can we interpret the def?

Hypothesis Testing



→ Alice is in S but not S'

→ Based on **Output**, μ , How to test optimally for S or S' ?

Null Hypothesis

H_0

Database is S

Alternative Hypothesis

H_1

Database is S'

Null Hypothesis

H_0

Database is S

• Type I error (False Negative):

Reject H_0 (i.e., accept H_1) under $M(S)$

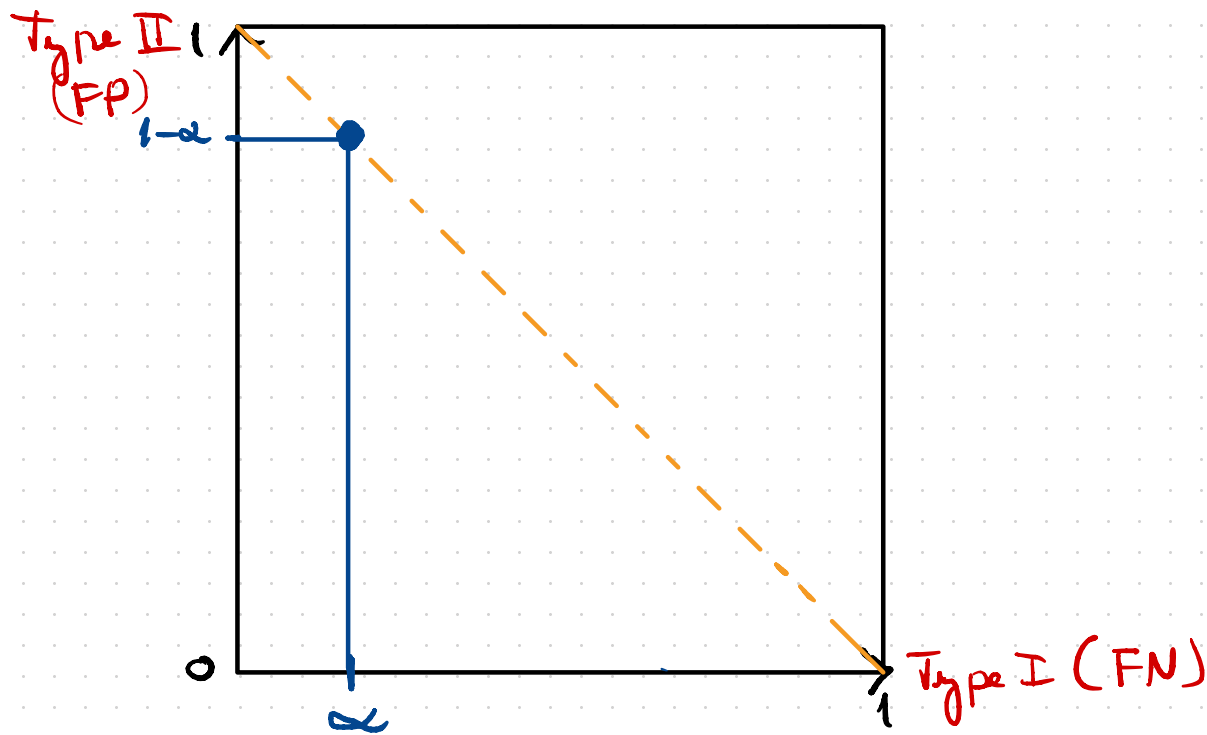
• Type II error (False Positive):

Accept H_0 (i.e., reject H_1) under $M(S')$

Alternative Hypothesis

H_1

Database is S'



• Random Guess

Fix a FN rate α . Then

- $\left\{ \begin{array}{l} \text{Predict } H_0 \text{ w/ probability } 1-\alpha \\ \text{Predict } H_1 \text{ w/ probability } \alpha \end{array} \right.$

General Prediction Rules:

ϕ : Outputs $\longrightarrow [0, 1]$

$\phi(x) \equiv$ Probability to pick H_1

• Type I error (FN): $\alpha_\phi = \mathbb{E}_{x \sim \mathcal{M}(s) \subset H_0} [\phi(x)]$

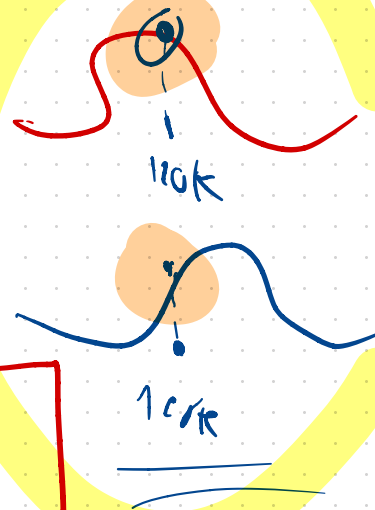
• Type II error (FP): $1 - \beta_\phi = \mathbb{E}_{x \sim \mathcal{M}(s')} [\phi(x)]$

Trade-Off function:

$$\left| \frac{p(x)}{q(x)} \geq 1 ? \right|$$

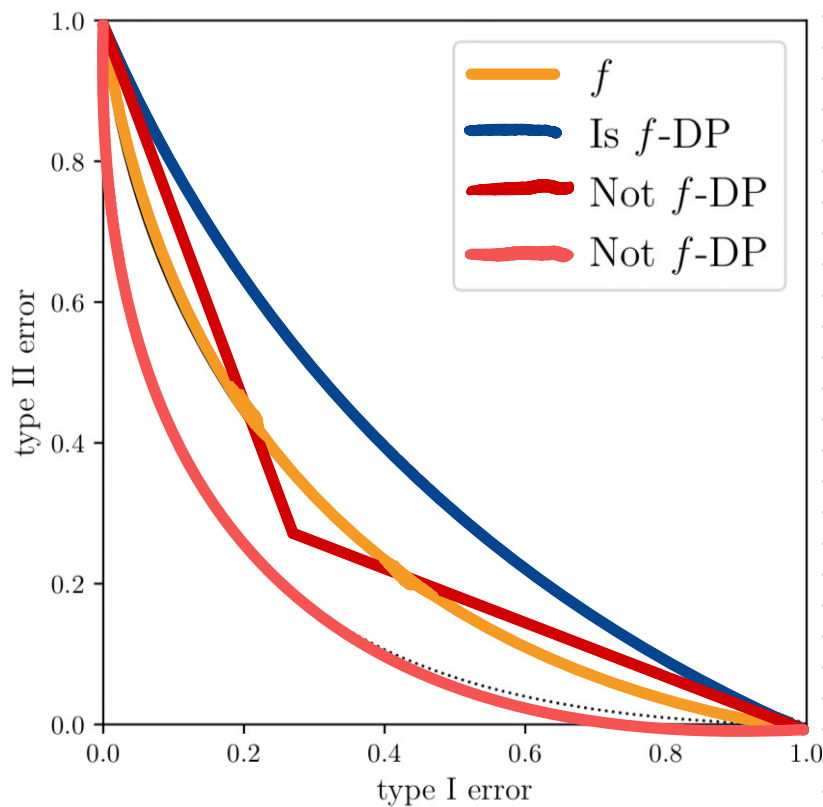
$$T(\mathcal{M}(s), \mathcal{M}(s'))(\alpha) = \inf_{\phi} \{ \beta_\phi : \alpha_\phi \leq \alpha \}$$

110k



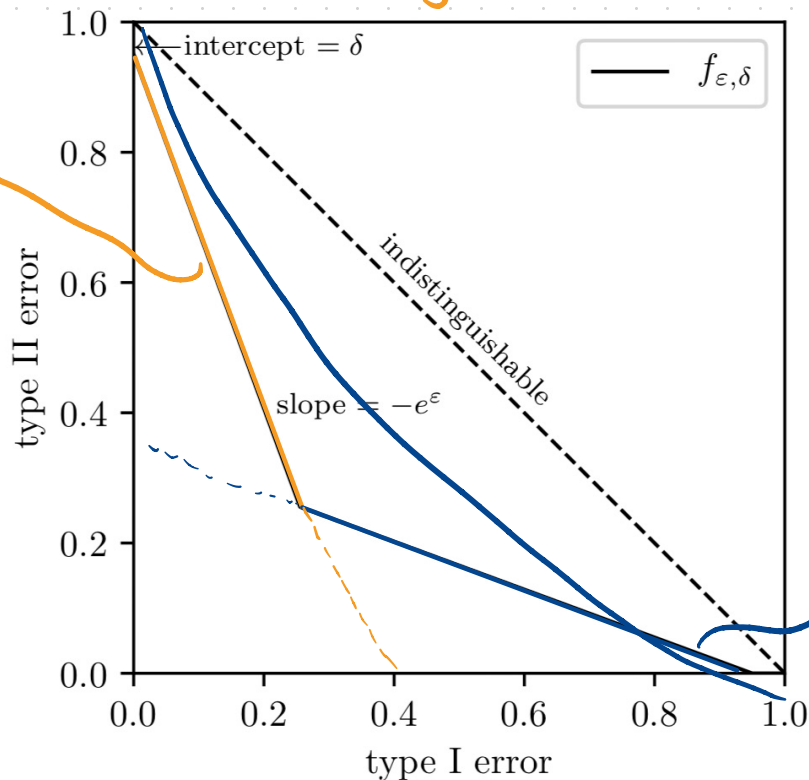
f -DP Definition: $T(M(s), M(s')) \geq \beta$

(Trade-off function)



Back to (ϵ, δ) -DP :

If $\mathcal{M}$ is (ϵ, δ) -DP, is it f -DP? For which f ?

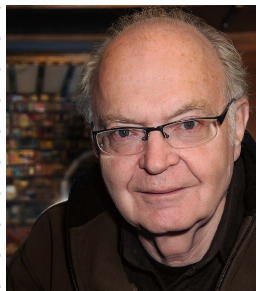


$1 - \delta - e^{-\epsilon} \cdot \alpha$

$f_{\epsilon, \delta}$

$\frac{1}{e^\epsilon}(1 - \delta - \alpha)$

What is (one) of the problems w/ (ϵ, δ) -DP?



"Premature optimization is the root of all evil"

- Summarizing DP in 2 parameters loses too much information

$$\left. \begin{array}{l} M_1 \text{ } (\epsilon_1, \delta_1)\text{-DP} \\ M_2 \text{ } (\epsilon_2, \delta_2)\text{-DP} \end{array} \right\} \Rightarrow (M_1, M_2) \text{ is } (\epsilon_1 + \epsilon_2, \delta_1 + \delta_2)\text{-DP}$$

NOT TIGHT!

→ Advanced Composition helps a bit

→ Divergence Approach $\left\{ \begin{array}{l} \text{Better/Exact composition} \\ \text{Not interpretable} \end{array} \right.$
(Rényi, ϵ -CDP, etc)

→ Post-Processing property for f-DP

Lemma 2.9 $T(\text{Proc}(P), \text{Proc}(Q)) \geq T(P, Q)$
↳ Data independent
Post-processing

Corollary: M is f-DP $\Rightarrow \text{Proc}(M(\cdot))$ is f-DP

→ "Optimality" of f -DP

Theorem 2.10 (Blackwell '50)

$\exists$ Proc s.t.

$$T(P, Q) \leq T(P', Q') \iff \begin{aligned} P' &= \text{Proc}(P) \\ Q' &= \text{Proc}(Q) \end{aligned}$$

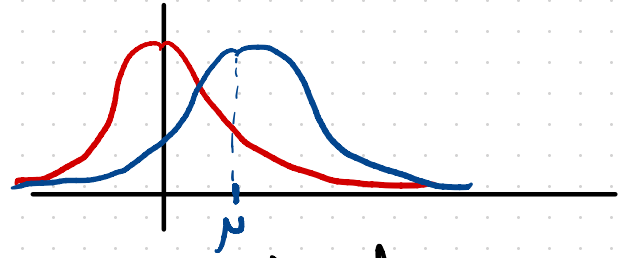
Corollary | Let $D(\cdot \parallel \cdot)$ be a "divergence" such that
 $D(\text{Proc}(P) \parallel \text{Proc}(Q)) \leq D(P, Q)$

Then $T(P', Q') \geq T(P, Q) \Rightarrow D(P' \parallel Q') \leq D(P \parallel Q)$

Corollary | $D(P \parallel Q) = \ell_D(T(P, Q))$

Gaussian Differential Privacy

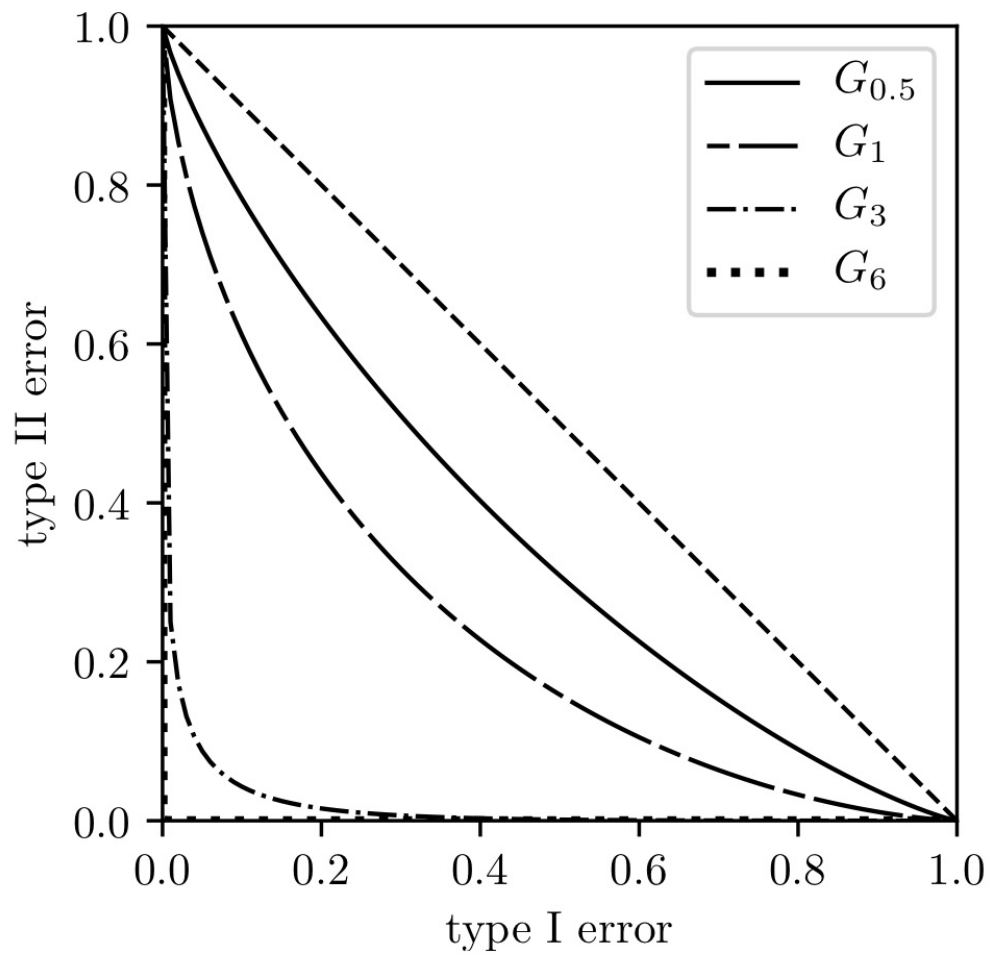
$$G_\mu := T(N(0,1), N(\mu,1))$$



Def: M is μ -Gaussian DP (μ -GDP) if
it is G_μ -DP, that is,

$$T(M(S), M(S')) \geq G_\mu$$

"As hard to tell apart as differing the 2 Gaussians"



Simple example using GDP

→ We want to privatize a statistic $q(s)$

↳ Sensitivity: $\Delta := \sup_{s, s'} |q(s) - q(s')|$

→ $M(s) := q(s) + Z$, $Z \sim N(0, \frac{\Delta^2}{\mu^2})$

Theorem 2.7 | M is μ -GDP

→ This is an **EXACT** characterization of the privacy of the Gaussian Mech:

$$G_\mu(\alpha) := \inf_{s, s'} T(M(s), M(s'))(\alpha)$$

Group Privacy

Thm 2.14 | $\mathcal{M}$ is f -DP. Then

$$T(\underbrace{S, S'}_{\substack{\text{differ by} \\ \underline{k} \text{ elements}}}) \geq 1 - \underbrace{(1-f)^{0/k}}$$

differ by
k elements

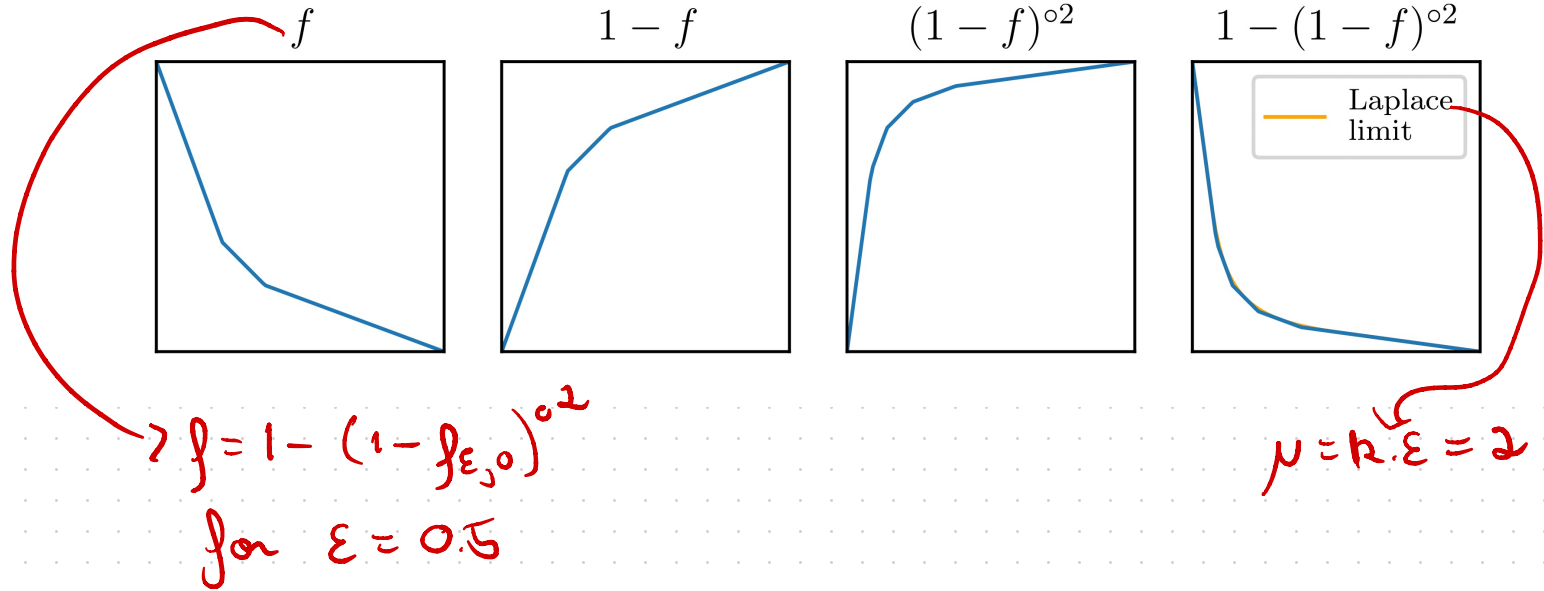
$$h^{0/k}(x) = \underbrace{h(h(h(\dots(h(x)\dots)))}_{k \text{ times}}$$

In particular, if $\mathcal{M}$ is ρ -GDP, then

$\mathcal{M}$ is $\underline{k\rho}$ -GDP for groups of size k

→ Asymptotic Behavior of pure DP

Prop 2.15 Fix $\mu > 0$, $\varepsilon = \frac{\mu}{k}$. Then
 $1 - (1 - f_{\varepsilon,0})^{\circ k} \xrightarrow{k \rightarrow +\infty} T(Lop(0,1), Lop(\mu,1))$



Composition

Def: $f = T(P, Q)$, $g = T(P', Q')$

$$\underline{f \otimes g := T(P \times P', Q \times Q')}$$

Thm 3.2 | M_i is f_i -DP for $i = 1, \dots, n$
 $\Rightarrow (M_1, \dots, M_n)$ is $f_1 \otimes \dots \otimes f_n$ -DP

• $G_{\mu_1} \otimes G_{\mu_2} \otimes \dots \otimes G_{\mu_n} = G_{\mu}$ for

$$\mu = \sqrt{\mu_1^2 + \dots + \mu_n^2}$$

Central Limit Theorem for f-IDP

General idea: For $f_1, \dots, f_n$ "nice", we have

$$f_1 \otimes f_2 \otimes \dots \otimes f_n \approx G_\mu$$

The formula for μ is hard to interpret. Let us look at $(\varepsilon, 0)$ -IDP!

Thm 3.6 | If $\sum_{i=1}^n \varepsilon_{ni}^2 \rightarrow \mu^2$, $\max_{1 \leq i \leq n} \varepsilon_{ni} \rightarrow 0$. Then

$$f_{\varepsilon_{n1}, 0} \otimes \dots \otimes f_{\varepsilon_{nn}, 0} \rightarrow G_\mu$$

Example: $\varepsilon_{ni} = \frac{1}{\sqrt{n}}$

Central Limit Theorem for (ϵ, δ) -DP

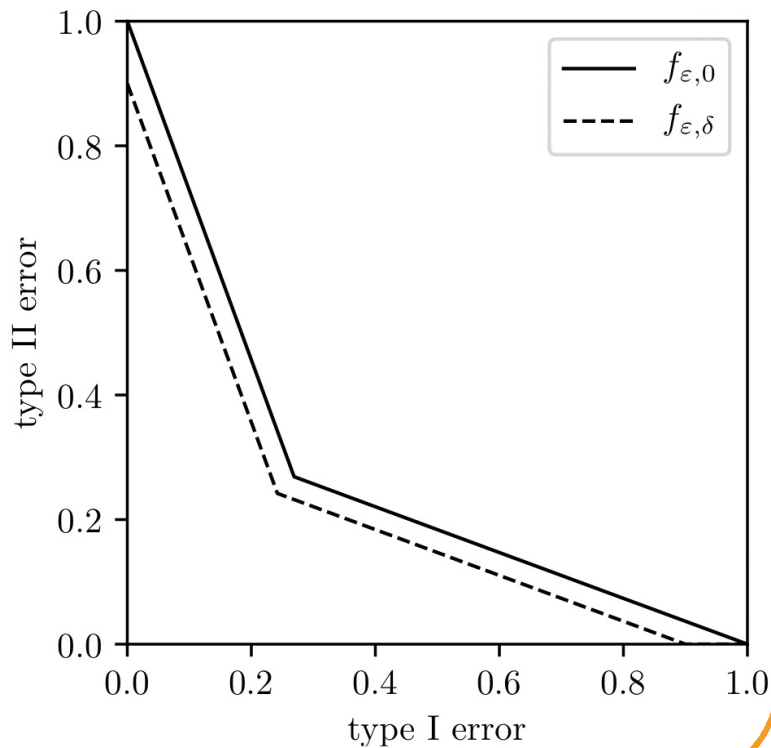
→ The last CLT result does not hold for $\delta > 0$

→ But $f_{\epsilon, \delta} = f_{\epsilon, 0} \otimes f_{0, \delta}$

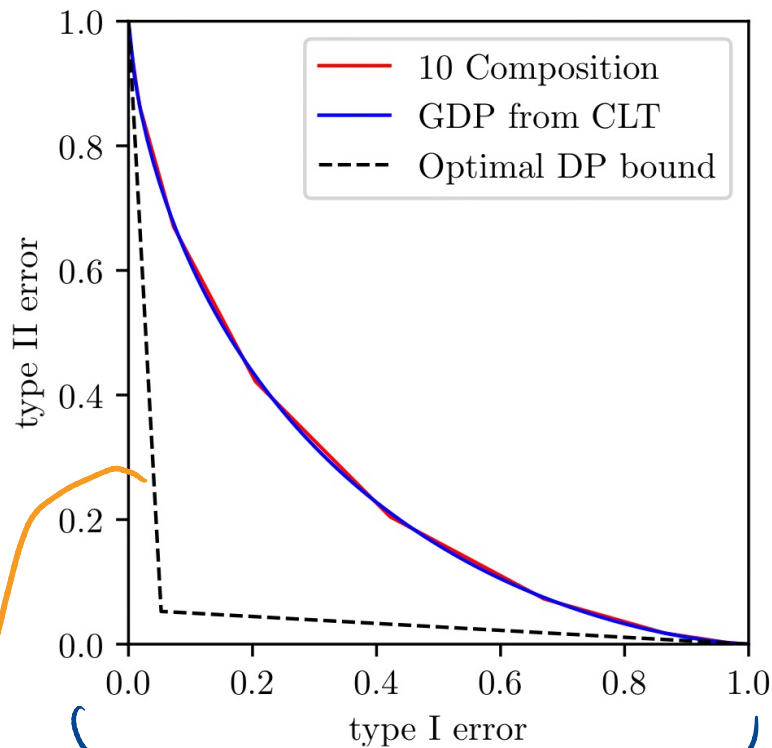
Claim If $\sum_{i=1}^n \delta_{m_i} \rightarrow \delta$ and $\max_{1 \leq i \leq n} \delta_{m_i} \rightarrow 0$
 $f_{0, \delta_{m_1}} \otimes \dots \otimes f_{0, \delta_{m_n}} \rightarrow f_{0, 1 - e^{-\delta}}$

> Ex: $\delta_{m_i} = \frac{\delta}{n}$

Intuition: Attack against $(\mu, \delta)^{\text{DP}}$ is as hard as distinguishing
 $N(0, 1) \times U[0, 1]$ and $N(\mu, 1) \times U[\delta, 1 + \delta]$



$f_{\epsilon,\delta}$
 $\omega/\epsilon \approx 2.81$
 and $\delta \approx 0.001$



Composition of $(1/\sqrt{10}, 0)$ -DP
 10 times

From f -DP to (ϵ, δ) -DP

